DA4022 — Course Project Proposal

How Much PENCIL Can You Afford? Compute-Allocation Laws for Link Prediction

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Type: Self-proposed, extending our paper presentation — *Plain Transformers are Surprisingly Powerful Link Predictors* (PENCIL, ICML 2026). Closest anchor: #2.

PENCIL matches heuristic-informed GNNs at link prediction using a plain Transformer over sampled subgraphs. Its accuracy, though, is bought with compute: Section 7 states that "its GPU compute and memory scale linearly with the number of candidate links," and the authors name resource-efficient variants as future work. Expressive power likewise rises with the sampling budget, so quality is capped by what one can afford *per edge*. Neither the cost of a prediction, nor the accuracy that cost buys, is ever reported as a first-class result.

We propose to supply that analysis. Production retrieval systems never score every candidate with an expensive model: a cheap stage shortlists candidates and the expensive model re-ranks the top *k*. PENCIL's authors suggest exactly this hybrid, citing ContextGNN, and never test it. We will build it — Adamic–Adar as the first stage, PENCIL as the re-ranker — then treat inference as a budget-allocation problem between *k*, the number of candidates re-ranked, and *s*, the sampling budget each one receives. Under a fixed budget, is it better to re-rank many candidates shallowly or few deeply? We will sweep both on cora, citeseer and ogbl-collab, measuring accuracy against latency and peak memory, map the accuracy-versus-cost Pareto frontier, and fit a rule for choosing (*k*, *s*) which we then test on a graph it was not fitted on.

Feasibility is already established: we have reproduced the paper's cora result — test MRR **44.00** against the published **42.23** — in 3.2 hours on a single free Kaggle T4, where the authors used four to eight A100-class GPUs. We hold the trained checkpoint, and the *k* axis needs no retraining, since the number of candidates re-ranked is purely an inference choice.

As this is a self-proposed project, we would be grateful for your approval. Should self-proposed projects not be permitted, we would like this registered under ***anchor 2 (task-dependent compute allocation laws)****, which this project directly addresses — it derives exactly such a law, for link prediction, on the inference budget.*